

# MINIGRAM CORE SYSTEM DOCUMENTATION

## ■■ CORE SYSTEM (ENGINE BRAIN)

FILES:

- main\_js/core.js
- main\_js/bundles/core.bundle.js

## ■ WHAT IS CORE SYSTEM?

CORE SYSTEM is the BRAIN of MINIGram.

It controls:

1. Global STATE management
2. Navigation system
3. App initialization hooks
4. Event binding (DOM ready)
5. Global function exports safety layer
6. Error handling system

Think of it like:

- React runtime core
- OR game engine main loop
- OR OS kernel (light version)

## ■ SYSTEM ARCHITECTURE FLOW

DOM READY →  
initNavigation() →  
  Button bindings →  
    loadPage() triggers →  
      UI system updates

## ■ GLOBAL STATE SYSTEM

```
window.STATE = {  
  FEED: [],  
  PAGE: 0,  
  LIMIT: 5,  
  LOADING: false,  
  END: false,  
  postChannel: null  
}
```

USE:

- FEED → stores posts data
- PAGE → pagination control

- LIMIT → items per load
- LOADING → prevents duplicate API calls
- END → stops infinite loading

BENEFIT:

- ✓ Prevents duplicate requests
- ✓ Centralized data control
- ✓ Feed optimization

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## ■ NAVIGATION SYSTEM

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initNavigation() handles:

Buttons:

- homeBtn → home page
- searchBtn → search page
- profileBtn → profile page
- reelsBtn → reels page
- createBtn → create post

SAFE DESIGN:

- Uses optional chaining (?.)
- Wrapped in try-catch
- No crash if element missing

BENEFIT:

- ✓ Crash-proof UI navigation
- ✓ Easy page switching
- ✓ Modular button system

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## ■ DOM READY HANDLER

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document.addEventListener("DOMContentLoaded")

ROLE:

- Ensures DOM is fully loaded
- Prevents null element errors
- Initializes navigation only

IMPORTANT FIX:

- initApp() removed
- Only ONE entry point exists

BENEFIT:

- ✓ No double initialization bugs
- ✓ Stable startup system

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## ■ SAFE GLOBAL EXPORT SYSTEM

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Exports functions only if not already defined:

window.loadPage  
window.createPost  
window.openChat

#### FEATURES:

- Prevents overwriting
- Avoids duplicate function bugs
- Safe modular integration

#### BENEFIT:

- ✓ Plugin-safe architecture
- ✓ No function conflicts
- ✓ Scalable system design

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### ■■ ERROR HANDLING SYSTEM

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window.addEventListener("error")

#### ROLE:

- Logs file errors
- Captures runtime crashes
- Debug support

#### BENEFIT:

- ✓ Easy debugging
- ✓ Crash tracking
- ✓ Development safety

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### ■ CORE SYSTEM ROLE IN MINIGram

CORE SYSTEM = ENGINE BRAIN

#### FLOW:

CORE →  
STATE →  
NAVIGATION →  
PAGE SYSTEM →  
UI SYSTEM →  
FEED SYSTEM

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### ■ BENEFITS SUMMARY

1. Central control system
2. Safe navigation layer
3. Global state management
4. Crash protection
5. Modular architecture
6. Scalable app design
7. Clean startup flow

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### ■ SIMPLE UNDERSTANDING

CORE SYSTEM = Controls everything behind UI  
It does NOT render UI  
It only manages logic + flow

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